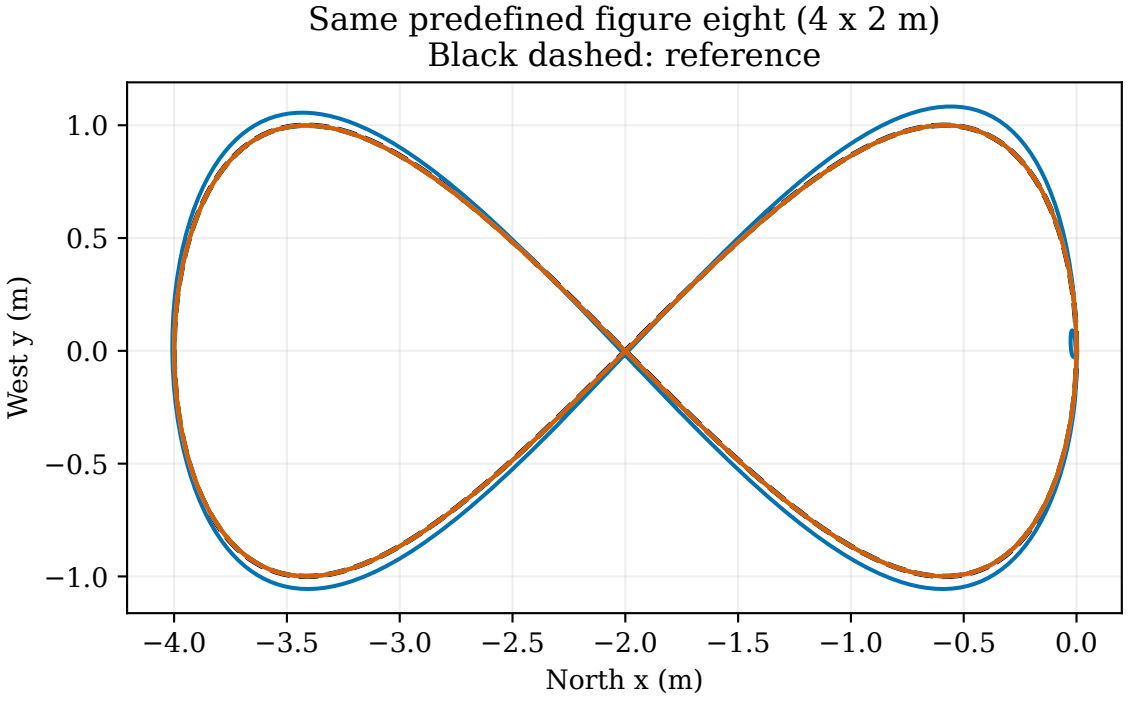
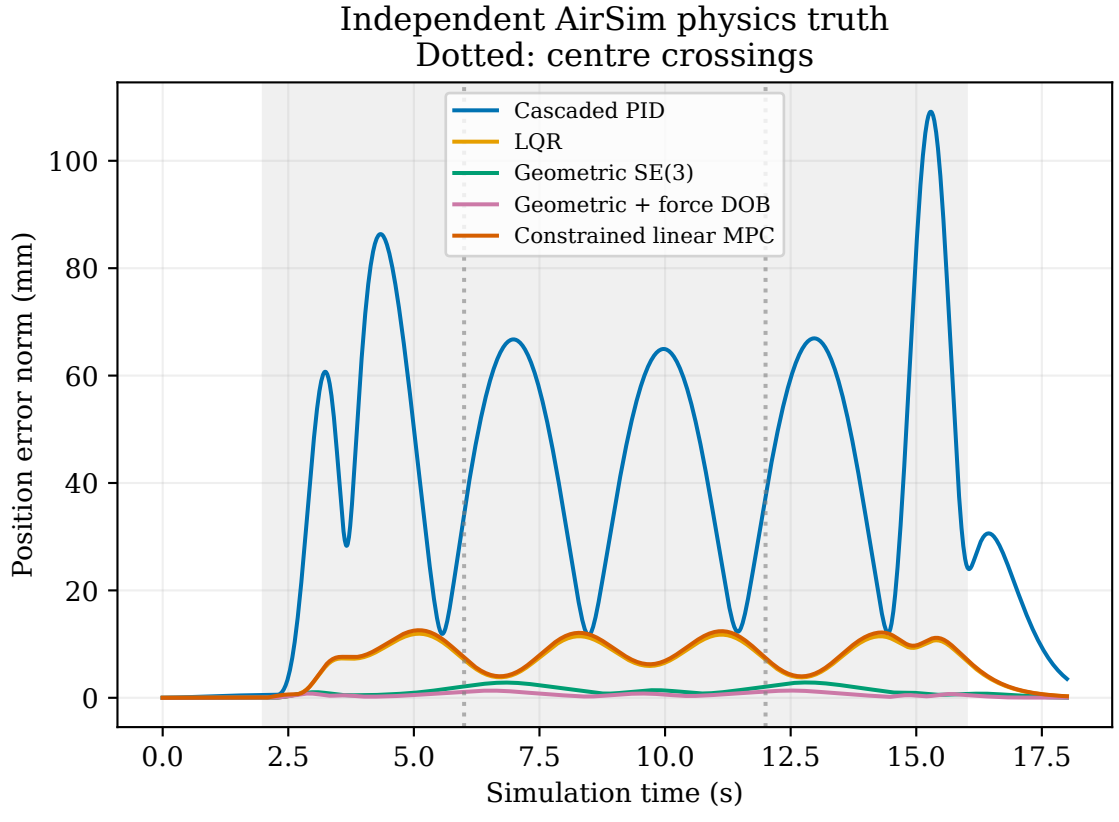

Same predefined figure eight (4 x 2 m)
Black dashed: reference
West y (m)
North x (m)
1.0
0.5
0.0
−0.5
−1.0
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0


Independent AirSim physics truth
Dotted: centre crossings
Cascaded PID
LQR
Geometric SE(3)
Geometric + force DOB
Constrained linear MPC
Position error norm (mm)
Simulation time (s)
100
80
60
40
20
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5